

```
"C:\Program Files\Python313\python.exe" C:\Users\Mahima
\OneDrive\Desktop\project\mahima.py
```

```
Customer ID          0
Age                  0
Gender               0
Item Purchased       0
Category             0
Purchase Amount (USD) 0
Location             0
Size                 0
Color                0
Season               0
Review Rating        0
Subscription Status  0
Shipping Type        0
Discount Applied     0
Promo Code Used      0
Previous Purchases   0
Payment Method       0
Frequency of Purchases 0
```

```
dtype: int64
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased',
      'category',
      'purchase_amount', 'location', 'size', 'color',
      'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

```
   age  age_group
0   55  middle-age
1   19  young adult
2   50  middle-age
3   21  young adult
4   45  middle-age
5   46  middle-age
6   63    senior
7   27  young adult
8   26  young adult
```

```

9    57    middle-age
      purchase_frequency_days frequency_of_purchases
0                14          Fortnightly
1                14          Fortnightly
2                 7           Weekly
3                 7           Weekly
4               365          Annually
5                 7           Weekly
6               90          Quarterly
7                 7           Weekly
8               365          Annually
9               90          Quarterly
      discount_applied promo_code_used
0                Yes           Yes
1                Yes           Yes
2                Yes           Yes
3                Yes           Yes
4                Yes           Yes
5                Yes           Yes
6                Yes           Yes
7                Yes           Yes
8                Yes           Yes
9                Yes           Yes
True
Index(['customer_id', 'age', 'gender', 'item_purchased',
      'category',
      'purchase_amount', 'location', 'size', 'color',
      'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'previous_purchases', 'payment_method',
      'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='str')

```

Process finished with exit code 0